Chapter 10
Selecting Measures for Research

Learning Objectives

10.1 Examine some of the key considerations when selecting a research measure

10.2 Examine the three avenues of choosing the appropriate measure in research

10.3 Report the need to be cognizant of related issues while choosing the applicable measures

10.4 Describe the implications of the terms brief measures, shortened forms, and use of single-item measures

10.5 Identify three explanations as to why the results obtained through multiple measures may vary

10.6 Examine convergent and discriminant validity

10.7 Review the need to ensure that the selected measure assesses the construct of interest

Systematic assessment is a fundamental across all of the sciences and accounts for enormous advances. Consider some familiar and unfamiliar examples in natural, biological, and social sciences:

- **In astronomy**, our earthbound and orbiting telescopes, more powerful than ever across the full wavelength spectrum, identify objects that are mysterious (never identified before) and less mysterious but new to us such as planets outside our solar system that appear to have the conditions close to those on earth and therefore in principle could be habitable. Habitable means that life might be on the planet and if so a new place for firmly establishing all the familiar fast-food restaurants.
- **In neuroscience**, neuroimaging of human and nonhuman animals has opened new ways of looking at the brain and generating and testing hypotheses about what process might be involved in learning, memory, cognition, social behavior, decision making, overeating, gambling, and more (e.g., Perkel, 2013; Sporns, 2010). The now very familiar methods (fMRI, PET) are early in the assessment methods and much finer-grained assessments of networks, circuits, and individual cell functioning are not on the horizon—many are here that can examine processes at the cellular, molecular, and now atomic level.
- **In genetics**, all sorts of assessment breakthroughs are evident and presented in the news as efforts are made to recover DNA and to have the possibility of cloning extinct species (e.g., Wooly mammoths, various dinosaurs, and Neanderthal people). More evident in everyday life is genetic mapping. In some applications, genetic testing can identify whether some individuals are at high risk for various diseases including various forms of cancer and heart disease but also forms of mental illness.
- **In anthropology**, tombs, lost civilizations, and buried cities and structures can be identified from space satellites (e.g., Schuetter et al., 2013). Once identified from space, the leads that could not be otherwise observed can be pursued directly on the ground to unearth previously hidden structures.
- **In ethology** or the study of nonhuman animal behavior in natural settings, we have learned about migration patterns, diet, social structure, and so much more through advances in assessment. Consider a recent example regarding our knowledge of cheetahs (I mean the animals and not individuals who lie on their income tax). Cheetahs are great hunters not merely or primarily because they are fast as we have always thought (Wilson et al., 2013). Outside of TV commercials, they do not use their great speed most of the time

when they hunt. Rather their hunting prowess comes from unusual agility in starting, stopping, and turning on a dime—more like a nickel.

- **In psychology**, fundamental advances in the psychology of learning, as reflected in the work of Ivan Pavlov (1849–1936) and B.F. Skinner (1904–1990), derived from novel and creative conceptualizations of the subject matter. Yet novel assessment methods used in the research (e.g., drops of saliva, rate of response) contributed greatly to the advances. The measures provided very precise objective measures that were responsive to training and a variety of experimental manipulations.

In each of the above examples, the news media understandably report the interesting tidbits of *what* we have learned. Yet in the background and rarely presented to the public at least is that the findings reflect breakthroughs in assessment.

The assessment methods allow more precise ways of revealing the subject matter of interest to the diverse sciences and often in the process reveal "new" phenomena—they were always there—we can now detect or see them or that look at "old" phenomena but at a new finer grained level of analysis.

All of this is to convey that assessment plays and has played a central, even if often unsung role, in scientific advances (see Greenwald, 2012). These advances are evident in psychological science and clinical psychology as a subarea within that. For example, neuroimaging and genetics have elaborated key topics of interest (e.g., aggression, violence, symptoms of depression, commonalities shared many psychiatric disorders, and changes when psychotherapy is effective).

Assessment in the context of our own studies has a much narrower and more concrete focus. We are going to test a conceptual view or hypothesis (e.g., about emotion regulation, empathy). Among the decisions for the study is selecting or developing the measures we will use.

This chapter focuses on overarching issues that influence or ought to influence measurement selection for research. We will cover validity and reliability of measures and how they influence measurement selection, the use of available measures and the development of measures when those are not too helpful, and many special issues and considerations such as the use of short forms, assessment biases, and others. Assessment is so critical to research and often is an afterthought or ancillary part. The two chapters identify key issues that can guide measurement selection.[1]

When we consider identifying measures for research, attention usually is drawn to the dependent measures or outcomes that will serve to evaluate the hypotheses. Yet, assessment entails all of the measures and their different purposes before, during, and at the end of the investigation:

- Assessments at the beginning of the study *before* the experimental manipulation may be used to evaluate demographic variables, to assess subject characteristics (e.g., exposure to abuse, self-control) that may relate to the outcome (i.e., moderators) to invoke subject selection criteria (inclusion and exclusion criteria), and to provide a base (e.g., pretest) of performance to pre- and post-measures to evaluate change.
- Assessment *during* an investigation (e.g., middle) might be used to check to see if participants really experienced the experimental manipulation or to assess intervening or mediating processes (e.g., emotion regulation strategies, alliance in therapy, cognitions or motivational states, and neurobiological processes [i.e., mediators and mechanisms]) that might explain or correlate with the changes over time.
- Assessment at the *end* of the investigation includes the dependent measures, i.e., those measures expected to reflect change. Also, at this point participants may be asked to reflect on features of the study, what they remember, know, and so on. Sometimes whether they experienced the experimental manipulation to which they were exposed is assessed at this time.

I have highlighted three temporal points (pre, mid, post) where assessments are routinely used, but of course this is a guide with exceptions you already know.

For example, longitudinal investigations assess participants over years and decades. Also, single-case designs assess behavior continually throughout the investigation. Other distinctions of when and how assessments are administered could be made as well.

But the key point is that assessment is central to different facets of the study at different points in time and with many different purposes. Measurement selection is critical.

Assessment is so critical to any study that it is important to provide guidelines and tools for selecting and perhaps avoiding certain measures. The emphasis will be on the dependent measures used for a study, but the key points pertain to assessment and selection of measures more generally.

10.1: Key Considerations in Selecting Measures

10.1 Examine some of the key considerations when selecting a research measure

The selection of measures for research is based on several considerations related to the construct validity of the measure, psychometric properties, and sensitivity of the measure to

the changes or differences predicted by the hypotheses. In a sense, we are "auditioning" possible measures for use in our study. It is not enough that a given measure appeared in a prior study, although that could be pivotal (e.g., for replication of one's own or prior work). We want to invoke several considerations for including the measure.

10.1.1: Construct Validity

As a general rule, in our research usually we are not interested in measures. An exception of course is when we are developing or evaluating a new scale or measure or making a career out of a particular measure that we believe is important and want the rest of the world to use. In the usual case, we are interested in *constructs* or the concepts that these measures reflect. As obvious as this sounds, it is critical to bear in mind, because measures often are selected without sufficient attention to the extent to which they actually measure the construct or facet of the construct of interest to us. For example, there are many measures of stress or social support, two constructs often studied in clinical, counseling, and health research. But the measures are not interchangeable, do not invariably examine the same aspects of stress or social support, and may be quite different in their utility and relevance based on the facets of the constructs they emphasize. Also, investigators occasionally make up a couple of items for their study that are considered to measure support and provide no data to even suggest support of what is being measured.

Consider an example. Social support, stress, and clinical dysfunction are topics that encompass many areas of research (e.g., diagnoses, cross-cultural studies) in clinical psychology (e.g., Maulik, Eaton, & Bradshaw, 2010). At a clinic where I work, parents of aggressive and antisocial children complete a brief measure of social support that has been in use for some time (Aneshensel & Stone, 1982). The measure includes three scales (items):

- How often they have received support from someone (in the past 2 months)?
- How many supportive relatives are in their lives?
- How many supportive friends are in their lives?

The theory underlying the measure is that in times of stress, individuals seek social support and that support can buffer (mute, mitigate) stress. For purposes of my comments here, consider the three scales that all measure social support. In our work, the inter-correlations of the scales range from r (N = 300) = .22 to .36—all significant (p<. 001) but all rather small.[2] The overlap (that they are correlated) suggests they measure a similar construct, but the fact that they are so low means they are not interchangeable and findings obtained for one of the scales may not be obtained for another. Of course, there are many more social support scales in the world and bringing them all into the discussion, if data were available, would underscore the point. That is, all measures of a construct may not be highly related and, as the example noted previously, all items within a measure of a construct may not be highly related. That is not only or merely an assessment problem—it is also related to how constructs behave and how we as humans behave.

Let us consider a construct: love of one's significant other or partner. We may truly love our partner but of the 10 (let us say 10 just for discussion sake) indices or behaviors, thoughts, and feelings that are measures of true love, a given person may be only good at 4 of them and mediocre at 3 others, and would not recognize the remaining 3 if they splashed on his or her windshield while driving on a sunny day. In other words, one can be truly in love but not be high (or score highly) on all of the measures (indices). Like what we know from everyday life, one can be athletic or a great athlete without being good at all sports. What this means for research. Look carefully at a measure you are considering. Does it "get at" or capture those facets of the construct you have in mind, i.e., in relation to hypothesis? Are there other measures to be added to the assessment to more thoroughly represent the construct?

As a place to begin, the initial criterion for selecting a measure is evidence that the measure assesses the construct of interest.

> In assessment, the term *construct validity is used to refer generally to the extent to which the measure assesses the domain, trait, or characteristic of interest* (Cronbach & Meehl, 1955).

10.1.2: More Information on Construct Validity

Construct validity has been used throughout this text to refer to a type of experimental validity that relates to the interpretation of the basis for the effect of the experimental manipulation. In the context of assessment, the interpretation of the measure is at issue, namely, to what extent does the construct underlying the measure serve as the basis for interpretation of the measure?

> In assessment, *construct validity refers to the link between the concept behind the measure and research that attests to the utility of the construct in explaining the findings*.

Stated another way, construct validity refers broadly to the pattern of findings and their integration that bears on the interpretation of the measure.

> Construct validity does not reduce to a correlation between measures or the measure and some other criterion. Rather, it involves the accumulation of evidence from diverse sources.

And that can include many different types of validity, as presented later. In addition, how does the measure relate to other constructs and is that in keeping or consistent with what one might expect from the construct we are measuring?

What do you think?

Fundamentally, construct validity relates to one's conceptual model of the measure, what it assesses, how the measure reflects the underlying construct, and how that relates to other domains of functioning (MacKenzie, Podsakoff, & Podsakoff, 2011). Construct validity involves a broad theory of how a concept (e.g., empathy, kindness, volatile personality) "behaves" and therefore what the measure ought to show in many different contexts beyond any single study.

In a given study, the investigator may be interested in measuring "stress" or "emotional distress." There should be some initial assurance that the measure actually reflects the construct. The assurance comes from accumulated evidence that findings are consistent with this construct and findings that another construct that might be related is not very plausible. In other words, we consider a mini-theory about the construct and measure. If I have a measure of stress, individuals with these characteristics or in these situations (e.g., immediately after a national disaster, during final exam week, learning of a disastrous diagnosis) ought to score more highly than individuals not undergoing any of these. This line of thinking continues—stress as a construct would be manifest in what ways? Construct validity is enhanced with supportive evidence showing that these predictions hold up in multiple contexts and that "stress" is a parsimonious and plausible explanation of what is actually measured by the scale.

10.1.3: Reasons for Carefully Selecting Measures

From a more practical standpoint, when one selects measures for a research project, it is useful to go beyond merely saying, "Well one or two other studies used this measure so it must be fine." Skepticism but not cynicism is a key to methodology. Ask, "What exactly has been shown to relate to the measure? What is the evidence that this measure I am considering captures or encompasses the characteristic I wish to assess?" It is easy to be enticed by many available measures into the assumption that they assess a particular construct. Be strong because it is easy to be tricked in selecting measures and thinking you have what you want for your study. Likewise, when reading the results of the study, look closely at the measures that were used. Here are some reasons why.

First, measures usually have names that reflect the construct the investigator *intended* the scale to measure (and, of course, often the name of the investigator as well). Unfamiliar (and fictitious) examples readily convey the sorts of measures that are available such as the Lipshitz Depression Inventory, Stop-Following-Me Scale of Paranoia, or the Kazdin I'll-Bet-You-Won't Jump Measure of Risk-Taking. The names of various measures may be based on supporting evidence that a particular characteristic or construct in fact is assessed (i.e., construct validity) or merely reflect what the originator of the measure had in mind without the requisite evidence.

> Construct validity requires the accumulation of studies to establish what the measure assesses and whether performance on the measure across diverse circumstances would be expected in light of the construct.

Second, one may read the items and nod in agreement that they in fact are measuring what you want to measure. We will return to this notion of face validity, i.e., whether a measure looks on the face of it that the items are related to the construct. Yet, what items measure can differ from what they seem to measure. For example, in clinical psychology going back decades there have been measures of psychopathology, i.e., how many symptoms of psychiatric disorder or psychological dysfunction one has and how severe they are. Items seem obvious as people are asked to endorse whether and to what extent they are anxious, depressed, hear voices, and so on. Yet construct validity was not so clear because another construct often seemed to be involved (confounded with) called *social desirability*, i.e., placing oneself in a favorable light. That is low symptom scores might reflect few symptoms but also could reflect a response set of trying to look good or better than one really is by not endorsing symptoms. (I will mention social desirability in more detail later because it can play a major role in assessment.) In short, of course look at the items, but do not assume that the content as described in fact is what is really measured by the item(s) or would relate to other well-established measures of the construct the items appear to measure.

Finally, many questionnaires and inventories have been evaluated through factor analysis, a set of statistical procedures designed to identify correlated items that cluster together. Names of the factors also suggest what is "really" measured.

Essentially, the factors (subsets of items) may be presented to assess different constructs or different facets of a single construct. Here too, one must be cautious because the connection between the name of the factor and what the items have been shown to measure (construct validity) is not always clear. Also, whether a scale, factor, or subscale with a given name is what the investigator means by the construct underlying the investigation is not automatic.

For example, one might develop a measure of love, but there are different types of love and different relations in which they are manifest. Is this measure of love the type the investigator has in mind, or is it distinguished from other types of love, or from liking a lot, from loyalty, from positive affect ("warm fuzzies") in general that is not necessarily love? As a construct validity exercise, ask questions the next time someone says, "I love you." Ask the person, "What

kind of love? What evidence can you offer that you do not just mean, 'like me a lot?' Maybe it is just physical attraction because I am drop-dead gorgeous?" (Methodologists always ask a lot of questions like these which is why most of their relationships are statistical rather than interpersonal.) These are construct validity questions and are not trivial. Needless to say, there should be some evidence that the measure selected for research in fact assesses the construct of interest. If there is no evidence available, some steps within the study should be taken to provide information on validity and other psychometric properties.

As a summary guide to construct validity, it is fair for one's advisor, mentor, or colleague to ask the question as you design your study, "what makes you think those measures assess the constructs you care about?" Similarly, if you are reading an article and the title suggests constructs (e.g., violence, aggression, kindness, moodiness) be sure to look closely at the measure(s) to see what was actually assessed how strong the evidence is supporting the connection between the measure and the construct. Construct validity is a guiding conceptual answer—the evidence from multiple studies that the construct behaves as expected and the measure is not likely to be explained by other constructs. How does one really identify if these latter criteria are met? The answer comes from the current evidence on the psychometric characteristics or properties about the measure, which coincidentally is the next topic.

10.1.4: Psychometric Characteristics

There are many steps for establishing or deciding whether a measure adequately assesses the construct of interest. These steps, broadly conceived, refer to how the performance on the measure relates to other domains of functioning (other measures) across a variety of circumstances. There are several characteristics of performance on the measure that contribute to the construct validity of the measure.

> *Psychometric characteristics is a broad term that refer to the reliability and validity evidence in behalf of a measure.*
>
> *Reliability generally refers to consistency of the scores obtained for the measure, including consistency among items of the measure (i.e., how the items relate to each other), consistency between different parts or alternate forms of the same measure, and consistency in performance on the measure over time (test–retest for a given group of subjects).*
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> *Validity refers to the content and whether the scores on the measure are shown to assess the domain of interest and encompasses the relation of performance on the measure to performance on other measures at the same time or in the future and to other criteria with which the scores on the measure would be expected to be related (e.g., school achievement, occupational status, psychiatric diagnosis).*

Any single definition of reliability and validity is hazardous because both are broad concepts and each has several subtypes. Also, over the years the different types of reliability and validity and their meanings and terminology have varied, so there are some inconsistencies (slight unreliability) in use of the terms reliability and validity(see Wasserman & Bracken, 2013; Watson, 2012).

10.1.5: More Information on Psychometric Characteristics

Table 10.1 presents major types of reliability and validity that are commonly referred to and of clear relevance in evaluating whether or the extent to which a measure is suitable for one's work or project. The terms in the table are useful to know because they are core concepts within assessment. Also, the concepts of reliability and validity sensitize the investigator to a range of considerations. In any given situation, a specific type of reliability and validity may or may not be relevant.

For example, high *test–retest reliability* (e.g., Pearson product moment correlation or *r* that is > .7 or .8) over a period of a few months might be expected for a measure designed to assess a stable characteristic (e.g., traits such as self-control, extroversion, or altruism) but not for more transient characteristic (e.g., a state such as irritated mood or perhaps anger). Similarly, a high *concurrent validity* correlation of the measure (e.g., of social support) with other indices of the same construct (e.g., frequent activities with friends, reliance on friends for help) would also provide support for the measure. Yet, a high correlation of that same measure (of social support) with indices of other constructs (e.g., intelligence, conscientiousness) might raise problems. This is why construct validity relates to "theory," i.e., one's conceptualization of how the construct relates to other constructs and also draws on multiple studies. Construct validity is not about one or two correlations but rather if performance on the measure across different situations "behaves" as one might be expected by the concept underlying the measure.

> Measures known to reflect the construct of interest, and to do so in a reliable and valid fashion, bolster the confidence to which the investigator is entitled when interpreting the results of the study.

In selecting a measure, it is important for the investigator to examine the available literature to identify the extent to which the scores on the measures have met the pertinent criteria for reliability and validity in ways that approximate the use in the present study. Many resources are available in print and on the Web to help obtain this information (e.g., Buros Institute, 2011; www.unl.edu/buros; ericae.net/testcol.htm). Also, merely typing in the name of the measure or "validity" or "reliability of" and then the name of the measure on a search engine (e.g., Google Scholar) is an easy and excellent place to start.

Table 10.1: Commonly Referred to Types of Reliability and Validity

Type	Definition/Concept
Reliability	
Test–Retest Reliability	The stability of test scores over time; the correlation of scores from one administration of the test with scores on the same instrument after a particular time interval has elapsed.
Alternative-Form Reliability	The correlation between different forms of the same measure when the items of the two forms are considered to represent the same population of items.
Internal Consistency	The degree of consistency or homogeneity of the items within a scale. Different reliability measures are used toward this end, such as split-half reliability, Kuder-Richardson 20 Formula, and coefficient alpha.
Interrater (or interscorer) Reliability	The extent to which different assessors, raters, or observers agree on the scores they provide when assessing, coding, or classifying subjects' performance. Different measures are used to evaluate agreement, such as percent agreement, Pearson product-moment correlation, and kappa.
Validity	
Construct Validity	A broad concept that refers to the extent to which the measure reflects the construct (concept, domain) of interest. Other types of validity and other evidence that elaborate the correlates of the measure are relevant to construct validity. Construct validity both on the theory underlying the measure (i.e., how that construct relates to multiple phenomena) and then concretely on the relation of a measure to other measures and domains of functioning of which the theory underlying the measure may be apart.
Content Validity	Evidence that the content of the items reflects the construct or domain of interest. The relation of the items to the concept underlying the measure. This is often evaluated by having experts in the area of the construct make judgments about the measure being relevant. This can also be evaluated statistically (e.g., factor analyses) to see how items go together and reflect content areas one would expect.
Concurrent Validity	The correlation of a measure with performance on another measure or criterion at the same point in time.
Predictive Validity	The correlation of a measure at one point in time with performance on another measure or criterion at some point in the future.
Criterion Validity	Correlation of a measure with some other criterion. This can encompass concurrent or predictive validity. In addition, the notion is occasionally used in relation to a specific and often dichotomous criterion when performance on the measure is evaluated in relation to disorders (e.g., depressed vs. nondepressed patients) or status (e.g., prisoners vs. nonprisoners). Does the measure correlate with the grouping variable and distinguish groups? Correlation (point-biserial), maximum likelihood estimates, and discriminant analyses are some of the ways to test for criterion validity.
Incremental Validity	This refers to whether a new measure or measure of a new construct adds to an existing measure or set of measures with regard to some outcome. That outcome might be in the present or future. Incremental validity is evident if the new measure adds significantly (statistically) and can be evaluated in multiple regression and discriminant analyses.
Face Validity	This refers to the extent to which a measure appears to assess the construct of interest. Not regarded as a formal type of validation or part of the psychometric development or evaluation of a measure. To say that a measure has face validity usually means no "real" (other types of validity are available). The value of face validity may stem from the likelihood that the measure is seen as a reasonable reflection of the domain of interest by persons who administer or who complete the test, but that alone does not attest to construct validity or any other type of validity.
Convergent Validity	The extent to which two measures that assess similar or related constructs correlate with each other. Convergent validity is supported if the measure correlates with other measures with which it is expected to correlate. The correlation between the measures is expected based on the overlap or relation of the constructs. Concurrent validity that takes on special meaning in relation to discriminant validity. Together these types of validity can help separate whether the correlations between measures are due to the relations of the constructs that are assessed or common method factors (all self-report measures) and their various combination. A multitrait-multimethod matrix (discussed later in the chapter) and various data analytic techniques are used.
Discriminant Validity aka divergent validity	The correlation between measures that are expected *not* to relate to each other or to assess dissimilar and unrelated constructs. The validity of a given measure is suggested if the measure shows little or no correlation with measures with which they are not expected to correlate. The absence of correlation is expected based on the separate and conceptually distinct constructs.

NOTE: The types of reliability and validity presented here refer to commonly used terms in test construction and validation. See For Further Reading for resources that elaborate various types of reliabilities and validity.

In addition to selecting measures with reliability and validity data in their behalf, it is important to report on these characteristics when preparing a written report and it is important to look for information about reliability and validity of the measures when reading a report. A review of articles (>950) across seven health and behavioral journals revealed that authors very frequently fail to report on the reliability or validity of the measures they have used in the study (Barry, Chaney, Piazza-Gardner, & Chavarria, 2014). This is not merely a reporting issue but can mean there is no clear assurance that the constructs the authors were discussing in fact were suitably assessed by the measures.

10.1.6: Sensitivity of the Measure

The measure ought to be sensitive enough to reflect the type and magnitude of change or group differences that the investigator is expecting.

Measurement sensitivity refers to the capacity of a measure to reflect systematic variation, change, or differences in response to an experimental manipulation, intervention, or different group composition (e.g., as in a case-control study).

For example, if a study compared the effects of mindfulness training versus no training to reduce anxiety among persons visiting a dentist, a relatively large difference (effect size) might be expected between these two conditions. One might expect performance on the measure to be able to reflect that difference. If two versions of mindfulness were compared, the difference on the measure, even if there were one, might be more difficult to detect. Whether an effect or difference is obtained is in part a function of whether the measure can reflect differences and change, but of course also a function of what is being studied and compared.

Whether and how sensitive a dependent measure is to change or to group differences is difficult to specify in advance of a study because it depends on other details of the study (e.g., what the manipulation is, how strong or large the differences expected between groups). A few general desirable characteristics of the dependent measure can be identified:

1. The dependent measure should permit a relatively large range of responses so that varying increments and decrements in performance can be detected. If a scale has a narrow range (scores can only span from 0 to 10), the ability of the measure to delineate different groups or conditions may be a problem. Alternatively perhaps there are 10 items and each of these is evaluated by the subject on a scale from 1 to 7; the total across many items (e.g., 10 items) could then have a potential maximum score of 70. There might be categorical, yes-no questions that are not easily placed on a continuum (e.g., have you ever been pregnant, do you own a dog, and don't you just love methodology?). Here, the number of yeses across several items might be summed, on the assumption they represent the same construct of interest. General rule: we want our measures to have a healthy range from some lower score to some higher score so that groups, conditions, and people can be separated or differentiated by the measure. Single-item measures (that sometimes are used to assess a construct) and very short forms (often used to save time) of larger measures have a number of risks and problems (mentioned later) but one is that they do not provide a large range to discriminate among subjects exposed to different conditions or who show different levels of the construct of interest.
2. If participants score at the extremes of the distribution at pretest, this, of course, will only allow the investigator to detect varying degrees of change in the opposite direction at postassessment. If it is necessary to be able to detect change in only one direction, as might be the case in studies designed to compare two treatments both known to be effective, then the measure need not allow for bi-directional changes. In such a treatment study, individuals may be screened because they are extreme (e.g., high levels of posttraumatic stress disorder symptoms) and we expect and hope they get better from some effective intervention (but also probably through statistical regression). Yet, as a general rule, allow for bi-directional changes if possible or of possible relevance to the hypotheses. Even in a study screening for extreme scores, the experimental manipulation may have an opposite of the intended effects, at least for some of the participants. Assessing and evaluating these changes can be very important. In general, there should be some assurance in advance of the experimental manipulation that ceiling or floor effects will not be a limitation that could interfere with detecting differences among various experimental and control conditions. These effects restrict the spread of scores and could make the measure insensitive to real differences that exist.

Psychometric data for the measure and the possibility of a wide range for scores to vary are important, but it is also useful for the investigator to ponder the items a bit.

Often scales are used without really looking at the items carefully to see if it is reasonable to expect scores on the items to reflect change for a given group or differences between groups. Also, scrutiny of the items may lead to hypotheses about some portions of the scale (e.g., subscales, factors) that might be more sensitive to group differences than others and that may provide a more direct or specific test of the hypotheses. As the investigator ponders the contents of a scale, he or she may begin to think of alternative or additional measures to better test or elaborate the construct.

Overall, the sensitivity of a measure in an investigation should be assured the best one can prior to conducting the study. If a body of literature already shows the sensitivity of the measure to the manipulation or intervention or for group comparisons of interest, then preliminary work on this issue can be avoided. Many studies are conducted that closely build on or redress some ambiguity of prior research, and the evidence from prior research may be quite useful and relevant. If such evidence from prior and closely related research is not available, preliminary work before the full investigation might evaluate whether different manipulations reflect change on the measure. A small pilot study (e.g., 10–20 cases, 5–10 in each of two groups) can provide preliminary information about whether the measure could yield group differences (because all or most scores are not at the ceiling or floor of the scores). It is important to know whether the measure could reflect the predicted relation between independent and dependent variables. If no relation were demonstrated between the

independent and dependent variables at the end of the investigation, it would be reassuring to know that the reason for this was not the insensitivity of the dependent measure. An alternative to pilot work is to include the measure with several others on an exploratory basis and explicitly acknowledge in the investigation that one purpose is to explore the relation of a new measure with those already available in the literature. This latter alternative is a full-scale investigation rather than just pilot work.

10.1.7: Diversity and Multicultural Relevance of the Measure

There is another type of "sensitivity" that is required of measurement beyond the type just discussed. This has to do with whether or the extent to which measures used in a study are appropriate for, reliable, and valid for different groups within the study. Consider briefly background for this generally neglected consideration.

The population within the United States has changed and is changing markedly to reflect increased cultural pluralism and diversity. Currently, minority groups comprise 37% of the U.S. population, and this is projected to increase to comprise 57% by 2060 (~50 years) (United States Census Bureau, 2012). Currently, Hispanic American and Asian American groups are expected to increase from 17% and 5%, respectively, now to 31% and 8% in 2060. African Americans are projected to comprise roughly the same proportion of the population (14% currently and 15% in 2060). Non-Hispanic White Americans are projected to decline from the current 63% of the U.S. population to 43% by 2060. And still small in percentage but the fastest growing category is multiracial with African American-European Caucasian and Asian-European Caucasian being the two fastest growing groups within this category.

> The growing number of ethnic "minorities" is a critical point of departure for all human-related sciences. Certainly, we want psychological science to be relevant to the diversity of our culture and of the world cultures too.

Apart from the numbers, we know more now about the critical importance of culture. Culture and ethnic identity can be reflected in fundamental psychological processes (e.g., memory, perception, decision making). Also, central topics within clinical psychology such as rates and symptom patterns of psychiatric disorders, risk and protective factors, seeking of and response to psychological treatment, and merely to mention a few topics are influenced, sometimes greatly, by culture and ethnicity (e.g., Paniagua & Yamada, 2013). The centrality of culture and ethnicity has been recognized nationally as reflected in U.S. Surgeon General's Report, which noted that culture identities "affect all aspects of mental health and illness, including the types of stresses they confront, whether they seek help, what types of help they seek, what symptoms and concerns they bring to clinical attention, and what types of coping styles and social supports they possess. Likewise, the cultures of clinicians and service systems influence the nature of mental health services" (e.g., Satcher, 2001, v).

Increasing attention has been accorded culture and ethnic diversity in clinical domains. Prominent among the acknowledgment in the mid-1990s to evaluate the role of culture in the context of psychiatric diagnosis was a Cultural Formulation Model (see Lewis-Fernández & Díaz, 2002; Mezzich, 1995). The Cultural Formulation Model was devised to recognize, consider, and assess five components:

- Assessing cultural identity
- Cultural explanations of the illness
- Cultural factors related to the psychosocial environment and levels of functioning
- Cultural elements of the clinician–patient relationship
- Overall impact of culture on diagnosis and care

The focus on diagnosis in cultural context is critical in its own right. As one illustration, diagnosis of serious psychopathology (e.g., psychosis) is more likely when cultural factors are not taken into account (Adeponle, Thombs, Groleau, Jarvis, & Kirmayer, 2012). Re-diagnoses recognizing cultural issues and the context of symptom presentation change the diagnoses individuals receive. The overall point is recognizing that culture is not merely a little moderator but can affect such weighty topics as prevalence of disorders and response to treatment.

10.1.8: Core Features of Ethnicity, Culture, and Diversity

Ethnicity, culture, and diversity more generally are core features of what we study. I have highlighted three points:

1. Changing demographics of cultural and ethnic groups in the United States
2. Role of culture and ethnicity in psychological processes
3. Acknowledgment in many areas of clinical research including but well beyond psychiatric diagnosis

Traditionally, research on ethnicity and diversity within the United States and cross-cultural research as part of international studies have served as two areas not well connected. Each area begins with the view that culture can moderate many findings and that understanding differences and similarities of different groups and subgroups is a point of departure rather than an afterthought. In light of the importance of culture, it is essential to draw some of the implications for assessment.

First, culture and ethnicity include critical components that can influence the inferences we draw from measures.

A given measure (e.g., Beck Depression Inventory, Positive and Negative Affect Scale) does not automatically provide the same information across cultural groups. Interpretation of the items, threshold for responding to a given item or symptom, and so on are likely to vary. We will discuss various types of reliability and validity later but it is important to note now that these characteristics are not properties of a scale or measure. Rather, they are the properties of scores obtained in a particular context.

> Critical to that context is cultural identity of the sample. It cannot be safely assumed, without supportive data within a study, that a given measure is equivalent for different cultures.

When one is including diverse samples in research, it is valuable to bring to bear data within the study that supports how the measures operate or behave (reliabilities, validities) for different groups. We already know that cultural identity can serve as a moderator and those points to evaluating the hypotheses by cultural identity. The assessment point is slightly different. Is a given measure assessing the same construct and with the same psychometric properties for the different cultural subgroups within the sample? Providing data on that within the study would be an excellent addition and arguable someday might even be required.

Second, much further work is needed on establishing the construct validity of measures for diverse ethnic and cultural groups.

Ensuring a measure is appropriate to different groups is not merely a matter of translating measures (and back translating to ensure the content is addressed). The language alone does not ensure that the measure is equivalent across cultural groups (see Leong & Kalibatseva, 2013). I mention this here because developing and evaluating measures as a function of culture and establishing similarities and differences in meaning and responses and across the full age spectrum are understudied areas of research.

In relation to a given study, I mentioned presentation of data on reliability and validity for the sample, especially if the sample departs from those used in prior research. The reason is that reliability and validity cannot be assumed in any new application. The recommendation is now expanded based on cultural considerations. If there are subsamples within a given study, report the assessment data in preliminary analyses to convey that reliabilities and validities (available to report within the study) operate in a similar way among subgroups.

10.1.9: General Comments

Selecting measures for a study is often relegated to looking at the existing literature and seeing what other investigators have used. This is kind of drive through restaurant shopping for measures to get the meal and measurement selection over with. I noted this merely to insert pause in selecting the measure for a given study.

Measures that have been used frequently and appear to show the effects of interventions or group comparisons by other investigators continue to be used frequently as new investigations are designed:

- On the one hand, using a common or consistent set of measures drawn from the literature has the advantage of permitting comparison of results across studies. One can tell whether subjects were similar (e.g., in degree of depression) and whether the independent variable (e.g., mood induction) affects the dependent variable in roughly the same way (e.g., direction, magnitude). Common assessment methods across studies greatly facilitate such comparisons.
- On the other hand, much research is conducted in a tradition of weak or narrow assessment with little innovation to push or elaborate the limits of a construct. Precedence (used in a study that has been published) is a *de facto* criterion for measurement selection, but not one of the stronger criteria.

> As a quick guideline, ask a fellow researcher (or yourself), why are you using *that* measure?

If the answer begins with a comment that others have used the measure, this conveys the potential problem. There are important considerations in selecting measures, and prior use of the measure by someone else may or may not be one of them. But it should not be the first reason unless one is trying to replicate the prior study and may not even be a good reason without reassurances that original investigator traversed the thought processes highlighted here. The reasons ought to be based on:

- Construct validity
- Psychometric characteristics of performance on the measure
- Sensitivity of the measure
- Cultural considerations as they pertain to the sample included in the study

> At the end of the study, suitability of the measure may emerge as a basis for criticizing the results. Indeed, when hearing the results of a study, it is almost always meaningful, cogent, and important to ask, "But how was *x* [e.g., construct or dependent variable] measured?"

(It is important to name the dependent variable—my experience is that people look quizzical if you actually say "*x*".) The reason is that there may be little generality of the findings from one measure of a construct to another measure of the same construct, as I illustrate later. Also, it may be that the measure was great (by some psychometric criterion), but it is arguable whether it assesses the construct of interest.

10.2: Using Available or Devising New Measures

10.2 Examine the three avenues of choosing the appropriate measure in research

In most cases, the investigator will use available measures and report psychometric characteristics reported for samples used in previous research. When measures of the construct of interest are simply not available, however, the investigator may make the decision to develop a new measure to address the questions that guide the study.

10.2.1: Using a Standardized Measure

Many measures are available in an area of research, and there is usually tacit agreement that certain types of measures, modalities of assessment, and specific instruments are important or central. For example, in studying adult depression, an investigator is likely to include a self-report measure (Beck Depression Inventory) and clinician rating scale (Hamilton Rating Scale for Depression). These modalities and these specific instruments have enjoyed widespread use, a feature that does not necessarily mean that the measures are flawless or free from ambiguity. These scales are considered to be the most well-researched within this area, and performance on the scales (e.g., scores that relate to the degree of depressive symptoms and correlates among these different levels of symptoms) is quite meaningful among investigators. The frequent use of the measures has fostered continued use, and researchers embarking on a new study (e.g., evaluating treatment for depression) usually include one or both of these in the broader assessment battery.

Another reason for using standardized measures, of course, is the amount of work that may have gone into the measures by other researchers. That work facilitates interpretation of the measure. For example, to assess intellectual functioning or psychopathology among adults, one might rely on the Wechsler Intelligence Tests (different tests from preschool through adulthood) and the Minnesota Multiphasic Personality Inventory (MMPI-2), respectively. Thousands and thousands of studies of these measures with diverse samples and diverse cultures facilitate their interpretation. Also, use of such well-studied measures lends credence that a new study assessed the construct of interest. Similarly, fMRI has been used as a neuroimaging technique for some time (beginning in the early 1990s), and much is known about its use. There may be controversy about what can be concluded in any given study (e.g., what can and cannot be concluded from brain activation) (e.g., Bandettini, 2012; Lee et al., 2010). Yet, there are now fairly widespread accepted and commonly used methods of fMRI and software, scoring, and data-evaluation techniques to interpret and display the findings. In all of these circumstances, the investigator need not worry about defending the measure or providing support for the construct validity in light of the prior work that has been completed.

> Yet, there can be a trade-off. Does the standardized measure assess the precise construct or aspect of the construct of interest?

If yes, that is wonderful. If no, this means one might have a wonderful measure of the wrong construct.

The prior comments argue for selecting a well-researched measure when possible and to be sure that the construct of key interest is measured by that. That is all well and good, but an overarching tenet of methodology and science is to be skeptical and to question (but try to be nice about it). So even if a measure is standard and well-researched, that does not free it from your skeptical evaluation. The reason is the potential weakness that comes from being a standard measure.

> Standard measures take on their own life in the sense that once they are used a few times, there is a snowball effect in the accumulation and accretion of other additional studies.

Soon the measure is used automatically in an assessment battery without scrutiny. Occasionally researchers come along and scrutinize the data (e.g., psychometric properties) for the measure and convey the slightly embarrassing news that the measure is not all that great.

Using the Hamilton's Rating Scale for Depression is an excellent example because of its common and widespread use in depression research, as I noted previously. Scrutiny of the reliability and validity data from 70 studies spanning over three decades revealed that the psychometric properties and individual items across many samples are not that great at all and key types of validity (e.g., convergent and discriminant) are lacking (Bagby, Ryder, Schuller, & Marshall, 2004). These authors properly ask the question whether this measure, considered as the "gold standard" for assessing depression, is really a "lead weight" and something we might abandon for better measures.

The key point is not about the Hamilton scale but rather about use of standardized measures. Sometimes the comfort they provide is just as weak as what a child gives as an excuse on the playground when caught doing something and replies, "Everyone else is doing it too." When that child grows up, one hopes she does not select measures for a study using that same rationale. There might be lesser used measures that are just as good or better and one might vary the measure or develop a measure better suited to one's hypotheses.

10.2.2: Varying the Use or Contents of an Existing Measure

A standardized measure of functioning, cognitive processes, personality, behavior, or some other domain may be available, although some facet of the investigator's interest may make that measure not quite appropriate. The measure may have been developed, established, and validated in a context different from that of the proposed study. For example, one might wish to assess a geriatric sample, but the measure of interest has been developed, evaluated, or standardized with young adults. Alternatively, the investigator may wish to assess a particular ethnic group whose language, culture, and experiences differ from those samples with whom the measure was developed. The reason for selecting the measure is that the method or content seems highly suitable for the investigator's purposes. Yet, the measure has not been used in this new way or validated in the new context.

As I mentioned and invariably important to bear in mind, reliability and validity are not characteristics embedded in a measure. Rather, psychometric properties are related to scores of the measure in a particular use (e.g., sample, context). It is useful to know that a particular measure has yielded adequate to good reliabilities and validities across many circumstances and that is one reason to consider use of that measure in closely related but new circumstances (e.g., slightly different application from prior studies). Yet, the new use cannot assume adequate reliability and validity. It becomes more difficult to persuade oneself as a researcher or as a reader of a research study that the measure was fine (reliable, valid) in this new use as that use departs from those conditions that have already been well-studied.

If one is applying a tried and true measure in a new use, it is very helpful to include within the study some effort to evaluate psychometric properties in this new use. The task is to provide evidence that scores on the measure behave in a way that parallels the more common and standard use of the measure. Evidence regarding reliability is very useful, but greater concerns are likely to be voiced in relation to validity of the measure in its new use.

> Evidence might include correlating scores on the measure in its new use with scores on other measures in the study or using the measure to delineate subgroups and showing that the findings resemble those obtained in studies when the original measure has been used as intended.

If one is preparing a manuscript (e.g., for publication or equivalent paper), before presenting the main findings, often it is useful to include in the Results section preliminary analyses that evaluate the measure in its new use with any psychometric (reliability, validity) data that could be brought to bear. It may be sufficient to show that the new use of the measure leads to predictable differences on the dependent measure, although this may vary as a function of the complexity of the predicted relations and the plausibility of alternative interpretations of the results on the measure. Yet it is even better to show that and some psychometric properties associated with the new use. In the general case, it is advisable within the study or as part of pilot work to provide additional evidence that the construct of interest is still measured in the new use of the measure and that the measure still enjoys adequate psychometric properties.

10.2.3: More Information on Varying the Use or Contents

Use of existing measures in novel ways is often preferable to creating entirely new measures because the available research on the existing measure (e.g., original factor structure, correlations with other measures, and psychometric characteristics from various samples) is still relevant for interpretation of the measure.

If an entirely new measure were created instead, none of this background information would be available. On the other hand, use of standardized measures in novel ways may be viewed and labeled by colleagues who review the research as inappropriate or beyond the intention of the founding fathers and mothers who devised the measure. There becomes a point at which applicability of the measure to new samples, populations, and circumstances is strained and the challenge is appropriate. For many colleagues, that point consists of any extension beyond the specific purposes for which the measure has been developed and standardized. Reasonable people differ on this point, but reasonable investigators (you and I of course) provide some validity data to ally the cogent concern that the novel use is inappropriate or difficult to interpret. The validity data are not merely intended to allay concerns of others; we want to be sure that more than anyone else that we are studying the phenomena of interest as intended.

Investigators often make slight variations in a standardized measure such as:

- Deleting a few items
- Rewording items
- Adding new items

The purpose is to make the measure better suited to the new population or application. For example, questions asking about suicide attempt or violent acts may be omitted in a study of a community sample because the base rates of these behaviors might be low and the items would be potentially upsetting and provocative in that context. Approval of the research (e.g., Institutional Review Board of a university) may even require deletion of items of a scale. The same measure in a clinic setting would include

the items given the goal of identifying the full range of symptoms and the expectation that such items may be required. Omission of one or two items is a minimal alteration of the scale, and the items usually can be interpreted as if the scale were the original, by making changes in subscale or total scores (e.g., by prorating missing items or imputing missing data for that item in another way). Yet, this is all a matter of opinion, which is why we provide data to show that the scale still behaves in the same way.

There are little data available on the extent to which investigators make minor alterations in measures and the impact of these changes on the findings. Yet, available evidence indicates that "standardized," well-used measures are not really as standard as we thought. An evaluation of research using the Hamilton Rating Scale for Depression found that there are at least 10 distinct versions of the scale in use based on variations in wording and the number of the items (Grundy, Lunnen, Lambert, Ashton, & Tovey, 1994). (And additional variations have been used since this study was completed [e.g., Bent-Hansen & Bech, 2011].) Moreover, each variation did not have suitable reliability or validity data in its behalf or the strength of data that characterized the original version of the scale. It is likely that many researchers have lost track of the original scale, because as Grundy and colleagues noted, citations to the scale in a given study often are mistaken, i.e., they refer to a different version from the one used in the study.

> In short, standardized tests are likely to be altered; it is important to provide data that the altered version is as meaningful and valid as the results from use of the original version.

As a more general rule, when one tinkers with the content or format of a measure, the requirements are similar. As a minimum, some evidence is needed within the study to show the measure continues to assess the construct of interest and behaves psychometrically in a defensible fashion. To the extent that the measure is altered and that the new use departs from the one for which the measure was standardized, stronger and more extensive validity data are likely to be demanded by the research community.

As an illustration, in the work of our research group, we have been interested in measuring hopelessness in children in part because of work on an inpatient service where many admissions were children with depression and/or suicidal attempt. Among the issues that make hopelessness interesting is the relation to depression and suicidal attempt and ideation in adults, a topic that continues to gather research (Hirsch, Visser, Chang, & Jeglic, 2012; Klonsky, Kotov, Bakst, Rabinowitz, & Bromet, 2012). Hopelessness, or negative expectations toward the future, has been reliably assessed in adults with a scale devised for that purpose (e.g., Beck, Weissman, Lester, & Trexler, 1974) and frequently used as the Beck Hopelessness Scale (e.g., Hirsch et al., 2012; Neufeld, O'Rourke, & Donnelly, 2010). In developing the scale for children, the items from the adult scale were altered to simplify the content and to be more relevant to children's lives. Clearly such changes are not minor modifications of a scale but lead to qualitative differences in focus and content. Hence it is not very reasonable to assume that the original validity evidence obtained with adults would apply to children. Initial studies were conducted to provide reliability and validity data. Internal consistency data and analyses of items paralleled the results obtained with the adults scale. In addition, the construct of hopelessness in children generated results similar to those obtained with adults. Initial studies of the Hopelessness Scale for Children found that hopelessness correlated positively with suicide ideation and attempt and depression and negatively with self-esteem (Kazdin, Rodgers, & Colbus, 1986; Kazdin, French, Unis, Esveldt-Dawson, & Sherick, 1983; Marciano & Kazdin, 1994). Such studies are promising insofar that they support the construct validity of the measure and are similar to findings with adults.

Even so one or a few studies are limited, perhaps especially so if they emanate from one research program. In the case of our research, the children were within a restricted age range 6–13 and were all inpatients from a psychiatric hospital. Also, a limited range of constructs and other measures were examined to evaluate validity of the scale. In short, the studies provide some, albeit very incomplete, evidence regarding the new scale and how it behaves. The task in developing a measure is not necessarily to complete the full set of validational steps. Once an investigator provides preliminary evidence and places the measure within the public domain, others may complete further studies that greatly extend research on construct validity and psychometric issues, as is the case for the measure of hopelessness in children (e.g., Fanaj, Poniku, Gashi, & Muja, 2012; Merry et al., 2012; Phillips, Randall, Peterson, Wilmoth, & Pickering, 2013).

10.2.4: Developing a New Measure

Sometimes measures of the construct of interest are simply not available. The investigator may wish to develop a new measure to address the questions that guide the study. Instrument development can serve as a program of research in itself and occupy a career. In most cases, investigators are not interested in developing or evaluating a measure with that in mind. Rather, the goal is to address a set of substantive questions and to conduct studies that measure the construct in a new way.

Developing a new measure is a weighty topic in its own right in light of advances in measurement theory and scale construction and is beyond the scope of this chapter (see Kaplan & Saccuzzo, 2013; Wasserman & Bracken, 2013).

In developing a new measure, some evidence is required, either in pilot work reported in the write-up of the study or as part of the study itself, which attests to the validity of the measure.

The steps extend beyond face validity, i.e., that the content of the items is reasonable or obvious. Various types of reliability and validity, as presented previously in Table 10.1, might be relevant. Particularly crucial would be evidence that supports the assertion that the measure assesses the construct of interest. Such evidence might be reflected in one or more of the following:

1. Differences between groups on the measure (e.g., older vs. younger, clinically referred vs. nonreferred cases) in ways that are consistent with the construct (criterion validity)
2. A pattern of correlations showing that the new measure behaves as predicted, i.e., evidence that the direction and magnitude of these correlations are consistent (e.g., low, moderate, high) with what would be predicted from the relation of the constructs encompassed by the new and more established measures (concurrent, predictive, or concurrent validity)
3. Evidence that the new measure is not highly correlated with standardized measure of some other, more established construct (e.g., intelligence, socioeconomic disadvantage, social desirability), which might suggest that the new construct is fairly well encompassed by or redundant with the other (more established) construct (and does not meet discriminant validity)
4. Evidence that over time, performance on the measure does or does not change depending on the nature of the construct (e.g., mood vs. character trait, test–retest reliability)

With the use of a new measure, evidence on one or more types of validity is a minimum required to argue that the construct of interest is encompassed by the measure. As noted in the discussion of altering a standardized measure, it is usually insufficient to add the measure to the study and to show that it reflects changes that are predicted. Within the study, separate and independent types of evidence are needed about the measure apart from or in addition to how the measure reflects change as a dependent measure. However, the persuasiveness of any particular demonstration on behalf of a new measure depends on a host of factors (e.g., complexity of any predictions and clarity of the findings).

As an example from our own work at a clinic I have mentioned, we have been interested in why families drop out of therapy prematurely, i.e., early and against advice of the therapist. Actually, I was not very interested in this, but the topic was forced on me in doing treatment outcome research with children refer for severe aggressive and antisocial behavior. Rates of attrition in child therapy are high in general (40–60%), but are particularly high among families of children with aggressive and antisocial behavior for reasons not yet clear. Some of the factors that predict dropping out are well studied (e.g., low socioeconomic status of the family, parent stress, single-parent families). Variables such as these are helpful in predicting who drops out but not very informative because they do not shed light on why someone drops out and hence what might be done to reduce dropping out.

We felt that for many families treatment itself raises barriers or obstacles that influence who drops out. We developed a measure, called the Barriers to Participation in Treatment Scale, based on our experiences with parents and obstacles they report (Kazdin, Holland, & Crowley, 1997; Kazdin, Holland, Crowley, & Breton, 1997). Meetings with therapists generated all we could think of from our cases (a few thousand families) of why they dropped out. We converted several of these reasons to specific items and piloted this to see how these items relate to each other. Finally, we selected 44 items that reflected stressor and obstacles that compete with treatment, treatment demands, perceived relevance of treatment, and relationship of the parent and therapist. We added 14 items to assess stressors unrelated to treatment (e.g., job stress, moving residences, and alcohol and drug problems).

The construct we wanted to measure (stressors associated with treatment) may be explained in part by stressors in the parents' lives that have nothing to do with treatment. The parent and therapist separately complete the scale; both versions are designed to capture parents' experience in coming to treatment.

The results of initial studies showed that scores on the measures predicted dropping out of treatment and other measures of participation in treatment (e.g., canceling appointments, not showing up), that scores on the measure were not explained by other more easily assessed variables that also contribute to dropping out (e.g., lower socioeconomic status, stress, and others), and that stressors associated with treatment are not explained by other stressors in the lives of the families.

What do we know from these initial studies?

Probably only that the measure is worth pursuing further. The results are consistent with the construct and provide preliminary support. All sorts of questions remain about the scale, content, and correlates and only a few of which have examined (e.g., Kazdin & Wassell, 2000; Kazdin & Whitley, 2006; Nock & Kazdin, 2005). Developing a new scale begins the path of validation completely anew, and initial studies are only a very first step. Many investigations and investigators are needed to extend the construct validity and applicability of the scale, refine its meaning, and clarify its utility (e.g., Smith, Linnemeyer, Scalise, & Hamilton, 2013; Williams, Domanico, Marques, Leblanc, & Turkheimer, 2012).

10.2.5: General Comments

The strength, specificity, and very likely the value or utility of the conclusions from a study depend on interpretation of what was measured and the meaning of performance on the measures. If extensive evidence is available for the construct validity of the measure, which is usually the case for standardized measures, the burden of interpretation is a reduced. The burden is never eliminated even here because psychological measures by their very nature raise manifold issues about construct validity, external validity, and potential response biases (e.g., social desirability was one already mentioned). Intelligence tests, for example, tend to be the most well-studied psychological instruments. At the same time, the tests are surrounded in controversy related to their interpretation and use, such as:

- What is really measured by the scales?
- Is this a special type and at that a narrow type of intelligence because it best predicts how well people do in school?
- How does this relate to other types of intelligence (e.g., problem solving) or other cognitive processes (e.g., decision making)?

As I have noted, if extensive evidence is not available for a measure or if the use of a well-studied measure is novel, it is valuable to include some information about the psychometric properties of the scale in the new use. Of course, sometimes one might develop a new measure and here of course much more extensive information is needed to suggest that the new measure is reliable and valid in some critical ways. Even though the goal of the study might be to test this or that hypothesis, it is useful to add to that a side light to provide data about some facets of reliability and validity.

10.3: Special Issues to Guide Measurement Selection

10.3 Report the need to be cognizant of related issues while choosing the applicable measures

There are several issues to be aware of and alert to when selecting measures. Perhaps the primary issue is what modality of assessment will be used, i.e., what types of assessment (e.g., questionnaires, psychobiological measures). Here I discuss issues that can address broader issues relevant to selection.

10.3.1: Awareness of Being Assessed: Measurement Reactivity

Measures most frequently used in research are presented to participants who are well aware that their performance is being assessed. Such measures are said to be *obtrusive to denote that participants are aware of the assessment procedures*. Obviously, participants know some facet of their personality or behavior is being assessed when they complete a self-report questionnaire or are placed into a somewhat contrived situation in which their behavior is observed.

> Awareness raises the prospect that performance on the measure is altered or influenced by this awareness. *If performance is altered by awareness of the measure, the assessment is said to be reactive.*

It is not necessarily the case that subjects' awareness (obtrusiveness) influences their performance (reactivity). Knowledge of the purposes of the measures and motivation of the subjects, and no doubt other influences (e.g., response sets), contribute to reactivity. A few problems can result from relying on measures when subjects are aware they are being assessed.

One problem that arises is that reactivity is a *method factor*, i.e., a characteristic of the measurement that may contribute to the results or scores on a measure. When two measures are administered, their correlation may be due in part because they were both obtrusive and reactive. Essentially, subjects may respond in a similar way across the two measures. Response set is a concept that captures one type of assessment bias that can emerge.

> *Response set or style in measurement refers to a systematic way of answering questions or responding to the measure that is separate from the construct of interest.*

The set or style is a systematic influence on how the individual answers the questions and can interfere with obtaining the true score on a measure. Table 10.2 summarizes four recognized response sets for easy reference. As noted there, the first one is an *acquiescence response set,* which is *a tendency for individuals to respond affirmatively (true or yes) to questionnaire items*. This does not mean that an individual high on this response set will answer all items in one way, but there is a tendency to agree that is systematic. What that means of course is that scores on a measure include one's standing on the construct (e.g., high in altruism) but also one's response set. This would occur in a simple situation where all the

Table 10.2: Response Sets that Can Influence Responding When Subjects Are Aware that They Are Being Assessed

Response Set	Defined
Acquiescence	A tendency for individuals to respond affirmatively (true or yes) to questionnaire items
Naysaying	Tendency for individuals to disagree and deny characteristics. This is the "other side" or opposite of acquiescence
Socially Desirable Responding	Tendency to respond to items in such a way as to place oneself in a positive (socially desirable) light
End Aversion Bias	A tendency to avoid extreme scores on an item (e.g., 1–7 scale) even if those extreme score accurately reflected the characteristic

items are coded in one direction so that a yes consistently means high in the characteristic (altruism). One can "fix" this so that agreeing for some items but disagreeing for others are signs of high altruism. That is, some items are "reverse scored" and hence worded in such a way that saying yes to most items does not lead to a systematic bias.

> A more well-investigated response set is socially desirable responding and is not so easily addressed. *A social desirability response set is where individuals tend to answer items in the direction of placing themselves in a positive (socially desirable) light*.

Here responses selected on a questionnaire or other measure where subjects are aware of assessment are in the direction of trying to provide a positive impression. Although we tend to think that only self-report questionnaires might be vulnerable to such biases, other types of measures (e.g., projective techniques and direct samples of behavior) have been known for some time to show such effects as well (Crowne & Marlowe, 1964). Here no matter how the items are worded (reverse scoring, requiring agreement or disagreement) the individuals tend to select alternatives that make them look good or make a socially positive impression. This is understandable of course. If the test score will influence being selected (e.g., for a job, for being connected to possible soul mates for a matching Website, for psychiatric hospitalization, for a team), one might be expected to hold back on admitting to socially checkered behaviors, characteristics, and tastes. Of course you might not want to risk mentioning any illegal behavior (e.g., for extra money you moonlight by selling drugs or that your streak of successful shoplifting without being caught has passed 100) or perfectly legal but low frequency behaviors (e.g., you used to be vegan but now pretty much you are a raw meat person; for years now your family Thanksgiving dinners are at drive-through restaurants). Socially desirable responding goes beyond psychological assessment. A concern in social media is that individuals may place something on their "page" that might be socially damning and could actually harm selection (e.g., admission to some program, receiving an award).

10.3.2: More Information on Awareness of Being Assessed

I mention response sets because they are a potential influence or bias in assessment when participants are aware that they are being assessed and that awareness can systematically influence their performance (see Podsakoff, MacKenzie, & Podsakoff, 2012). Systematic biases can operate even on measures that might seem relatively immune. For example, people often misestimate their height and weight when self-report is compared to actual measurement, and these differences vary as a function of age, sex, and culture (e.g., Burton, Brown, & Dobson, 2010; Spencer, Appleby, Davey, & Key, 2002). An early finding on the topic found that women tend to underestimate their weight much more than do men; men tend to overestimate their height much more than do women (Palta, Prineas, Berman, & Hannan, 1982). Perhaps this finding would fit in with impression management in light of cultural pressures on the different sexes.

In general, interpretation of psychological measures can be greatly enhanced by using multiple measures that vary in reactivity (e.g., one reactive, another not). For example, the construct may be operationalized by a self-report measure, but also by direct observation of performance out of the awareness of the participant or in a contrived laboratory situation where the purpose of the study and the assessment situation is ambiguous. If similar results are obtained across such measures, the investigator has greater assurance that conclusions are not restricted to some aspect of the assessment method or influenced by a particular response set.

10.3.3: Countering Limited Generality

The use of obtrusive and reactive measures may *limit generality* of research findings. The problem of reactivity of assessment can be elaborated by discussing external validity more directly. Because almost all psychological research with humans relies on subjects who know that their performance is being assessed, one can legitimately question whether the results would be evident if subjects did not know their performance was being assessed. We take for granted that how subjects respond to our questionnaires about stress, social support, and other key constructs really identify performance, perceptions, or feelings outside of our experiment.

> It is reasonable to assume that obtrusive measurement (e.g., questionnaires in the lab) is *correlated* with real-life (unobtrusive) indices of the constructs. Yet we have little idea of whether the correlation is very high.

The generalization question in relation to assessment is, "how does the subject respond when there is no special assessment situation (e.g., my study)?" Examining this question invariably improves the quality of the study.

Several solutions can minimize or even eliminate entirely the influence of subject awareness on performance. These solutions vary as a function of the specific method of assessment. With self-report questionnaires and rating scales, the instructions given to the participants often are designed to increase their candor and to decrease the influence of reactivity. **One tactic** is to tell the participants that their answers to the test items are anonymous and that their individual performance cannot be identified. Of course, in most investigations these claims are accurate, although the participants may not believe them. In other situations, instructions may be provided to minimize the

likelihood that participants will answer the items in a particular way. Subjects are more likely to respond candidly and less likely to place themselves in a socially desirable light if they believe they cannot be identified.

> Another strategy to minimize the influence of subject awareness on performance is to add *filler* or *buffer items* on a given measure. The filler items are provided to alter the appearance of the focus or to make the measure appear less provocative or intrusive. In the process, the true purpose of the measure, i.e., the construct of interest, is obscured.

For example, a self-report measure of various psychiatric symptoms, criminal activity, or sexual practices might be infused with items about interests, hobbies, and physical health. The participants are aware of the assessment procedures, but the filler items may obscure or diffuse the focus that would heighten reactive responding. The filler items may soften the impact of the measure, and the reactions that might otherwise be prompted. Of course, the success of such items to obscure or attenuate the emphasis is a matter of degree; adding a few buffer items (e.g., do you get colds a lot, have you ever collected coins or stamps as a hobby) to a newly developed Scale of Tendencies toward Extreme Terrorism may not help very much.

Another solution is to vary what participants are told about the task and how it should be performed. For example, the purpose of the test may be hidden or participants may be told that their test responses have no real bearing on their future and will not be used for or against them. Extremely bright or suspicious subjects recognize that statements like this reflect that in fact this information will be used for or against them. (This is sort of like a doctor saying to a child that, "this will not hurt!" One only learns through development such a statement often is a clear signal that something will be painful. In defense of our doctors and parents who say this will not hurt, anxiety and subjective experience of pain can be greater if one is expecting pain; turning off that expectancy is likely to reduce these [e.g., Ziv, Tomer, Defrin, & Hendler, 2010].)

> Alternatively, participants may be told to respond to the items very quickly. The purpose of "speed instructions" is to have subjects give little attention to what actually is measured and hence not deliberate about the content or purpose of the items.

These instructional ploys may or may not be plausible to the subjects, depending upon the circumstances of testing and the exact facets of personality or behavior that are assessed.

> The use of computers, mobile devices, and Web-based measurement in psychological assessment has implications for reducing the reactivity of assessment. Computers permit participants to answer questions directly by responding to items presented on a monitor or screen, often in the comfort of one's own home (or work place).

The questions are presented, and answers are recorded automatically without a human examiner. As mentioned previously, computerized assessment, when compared with the measure administered by an examiner, often yields more information about sensitive topics such as (e.g., alcohol consumption, sexual problems). In addition, respondents often report favorable attitudes toward computerized test administration. In short, although computerized assessment is obtrusive, it may be less reactive. Similarly, mobile devices assess functioning in everyday life. At random times of the day, an individual may be "beeped" to answer several questions about emotional states. The regular assessment on multiple occasions within a day and across days and assessment in everyday life may reduce biases in responding, a speculative comment yet to be tested.

When reactive procedures are used because of the unavailability of alternative assessment devices, one of the strategies that might be adopted is to encourage participants to respond honestly as possible. Although this may be naive when participants have a particular interest in their performance in light of some goal (e.g., job procurement, discharge from a hospital), the overall approach may be sound. In many cases, such as evaluation of progress in therapy, it is usually in the best interests of the client to respond as candidly and accurately as possible. In such cases, this message may be worth elaborating to the respondents to obtain samples of performance during assessment that are as representative of daily performance as the measures allow.

Assessment occasionally consists of direct observation of behavior over an extended period. With such measures, different solutions have been sought to decrease the influence of reactivity. For example, when behavior is directly observed in a naturalistic situation such as the home or at school, there may be a novelty effect and the early data may not represent daily performance.

> Usually the first few days are needed to individuals habituate to the observers. It is assumed that after a period of time, obtrusive assessment will become less reactive over time and exert little or no influence.

Whether performance under obtrusive and unobtrusive assessment conditions is similar requires empirical evaluation. Even under ideal conditions of administration, the fact that participants are aware that their behavior is to be assessed might affect generality of the results. Possibly the results of an experiment have little bearing on behavior outside of the reactive assessment procedures.

10.3.4: Use of Multiple Measures

As a general rule, more than one measure ought to be used in a given study to assess the (or each) construct of interest. It is rare that a single measure captures the construct completely

or well. There are important exceptions where one measure is viewed as *the* critical index of the construct of interest and there is relatively little or no ambiguity about the measure and the construct it reflects. For example, survival (i.e., not dying) is often used as a dependent measure in research on diseases and their treatment (e.g., heart disease and cancer). The measure (mortality) usually does not raise epistemological questions ("how do you know they were really dead?") or methodological challenges ("does 'not breathing' *really* get at the central features of the construct?" "What was the test–retest reliability of the measure?"). Of course, definitional questions arise when discussing life and death in the context of personal, social, and ethical issues (e.g., abortion, termination of life support systems) but not usually in the context of assessment for research purposes.

Multiple measures of a construct usually are advisable in research. Use of multiple measures may make the study more complex in many ways (e.g., more measures for the subject to complete, more data scoring and analyses, potential inconsistencies in the results).

Even so, the recommendation is based on two considerations:

1. Most constructs of interest (e.g., personality characteristic, clinical problem) are multifaceted; that is, they have several different components. No single measure is likely to capture these different components adequately. Consider the construct of depression. Some components of depression are based on self-report. Individuals *report* that they feel sad, worthless, and no longer are interested in activities that were previously pleasurable. In addition, there are *overt behavioral components*, such as reduced activity and social interaction and changes in eating (more or less eating). Similarly, *psychobiological components* include changes in sleep electroencephalogram activity. These different facets of depression may overlap, but they are not likely to be so highly related that one is redundant. Any evaluation of depression in, say, a test of treatment would be incomplete if change were merely demonstrated in one modality. Single measures might well be fine if the problem or focus is highly circumscribed (e.g., enuresis, isolated fears, and specific habit disorders), if the measure is one that the world views as rather definitive (e.g., death, DNA profile) or sufficient (e.g., heart rate, pulse), or the goal is to address a single facet of a problem (e.g., blood pressure as an outcome among hypertensive patients). However, in most research, multiple methods ought to be used whenever possible, at least for the core constructs of interest or that reflect the primary hypotheses.
2. Multiple measures of a construct are helpful to ensure that the results are not restricted to the construct as assessed by a particular method and measure. Performance on a given measure (e.g., score, level of the characteristic) is a function of both one's *standing on that characteristic* (e.g., level of self-esteem) and *the precise method in which assessment is conducted* (e.g., self-report questionnaire, one questionnaire vs. another). In other words, the measure itself can contribute to the findings and conclusions.

There is a natural tension in a given research project between representing a given construct well (by using multiple measures of that construct) and including multiple constructs with only one measure each in a study. It is not reasonable to require participants to complete onerously long assessment batteries. So multiple constructs in the study cannot each be represented by multiple measures. Also, invariably there are little pressures here and there from advisors, funding agencies, graduate students, or oneself to add a measure to get at one more construct.

The tension is between breadth of coverage (many different constructs) versus thoroughness in assessing a given construct.

The extremes could be represented by 5 measures of a single construct in a study or 10 measures of 10 different constructs. The former is fine but does not allow one to relate measure of the construct to other constructs; the latter is not likely to assess any construct particularly thoroughly.

In general, a compromise is useful to consider. Identify the main constructs of interest or the constructs that figure most prominently in the hypotheses. Let us call these the primary constructs because they guide the study. Here it would be useful to measure the construct(s) with more than one method of assessment. This will have all of the advantages mention before especially if the measures of the same construct include different methods of assessment (e.g., self-report, direct observation). Other constructs in the study may be secondary in that they are of interest and may be important for control purposes (e.g., as covariates) and if needed represent these with fewer or one measure each. This is obvious a compromise because representing a secondary construct with one is not ideal. Yet, this makes many studies feasible because of limits of what can be asked of the participants.

10.4: Brief Measures, Shortened Forms, and Use of Single-Item Measures

10.4 Describe the implications of the terms brief measures, shortened forms, and use of single-item measures

The term *assessment battery* refers to all of the measures that will be used in a given study.

Multiple measures in the battery can be extensive insofar as the study assesses multiple constructs (e.g., stress, social support, psychopathology, marital satisfaction, and emotion regulation), and some of the constructs may have two or more separate measures (e.g., self-report, other report, physiological response). While we are doing our arm-chair planning of what we want to assess, the assessment battery quickly burgeons to a long list of measures. Yet, the amount of time required by the participant to complete the measures can become prohibitively long (e.g., a few hours). Moreover, the participant may be required to complete the assessment battery on a few occasions at different points in the study (e.g., immediately before and after the experimental manipulation or at various intervals in a longitudinal study). Is there any possible relief for the poor participant? There are three assessment options. I will highlight each one briefly and then raise considerations and cautions.

10.4.1: Use of Brief Measures

This first option is important to mention to make a critical distinction for this discussion.

> A brief measure means what it says. The measure may have relatively few items and not take long to complete.

There is nothing special to note here because the requirements of a brief measure area all those discussed previously, i.e., establishing that the measure has the needed psychometric properties, in sensitive to change, and so on. Sometimes very brief measures can be quite useful after having demonstrated these characteristics.

For example, to identify symptoms of Generalized Anxiety Disorder, one scale was devised with 13 items and then dropped down to 7 items (Spitzer, Kroenke, Williams, & Lowe, 2006). The items were taken from the psychiatric diagnostic system in place at the time and the symptoms that defined the disorder (DSM, IV). Patients were asked how much a symptom bothered them during the past 2 weeks. Each symptom was scored on a 4-point scale from not at all, several days, more than half the days, and nearly every day. The validity of the 7-item version (called GAD-7) was supported in several ways (e.g., concurrent validity predicting lost days of work with higher levels of anxiety, patient status as evaluated by physicians, prediction of the full diagnosis and others) and was consistent with construct validity of the measure. This is a good example of a study with a brief measure that provides a careful evaluation of its validity.

There are issues that can emerge in brief measures, valid or not, and I shall take those up in discussion of considerations. However, as an initial point of departure, a standardized scale that is also brief raises no inherent problems. Such measures may be readily used because they have addressed the essential criteria in their development, because construct validity is defensible, and because of the added practical benefit of requiring less assessment time on the part of the participant than some longer version.

10.4.2: Use of Short or Shortened Forms

Brief measures can be considered as standardized scales that just happen to be brief. Short or shortened forms can be an entirely different matter. These are measures that are derived from and abbreviated versions of a longer standardized scale. Here too the investigator has as a goal in mind alleviating the burden of assessment and using a shortened version of a standardized scale.

As one example, consider the Symptom Checklist 90-Revised (SCL-90-R), a very widely used (internationally) self-report measure designed to assess psychiatric symptoms among adults (Derogatis & Unger, 2010). The 90 questions are presented in a true–false format and cover a wide range of symptoms domains (e.g., anxiety, psychoses, and depression). The symptom domains represent subscales, but research supports the total score as the more defensible measure because of the very high intercorrelations of the scales (rs > .9) and high internal consistency of the 90 items when considered as a single domain (by summing all of the items). The measure is used often in research but of course as only one of multiple measures. Many different shortened versions have been used.

For example, a recent report compared 3 shortened version of the scale (27, 18, and 14 items) with the full 90 item version among 2,727 patients with affective disorders (e.g., Prinz et al., 2013). Subjects completed the SCL-90 and from that full set of items, scores on the shortened versions were extracted. That is, patients did not complete four separate forms—they completed the full-length version, and smaller versions were made post-hoc by extracting subsets of items. The main findings were that psychometric properties (e.g., various forms of reliability and validity) were similar across the four versions of the measure. Also, all versions reflected changes in symptoms over time about equally well as patients completed the assessment on separate occasions. By and large the short form performed well insofar as showing results as obtained in the full 90-item version of the scale. There are many studies of SCL-90 and many short forms (e.g., in one study alone, 11 different short forms were evaluated [Müller, Postert, Beyer, Furniss, & Achtergarde, 2010]). In general, their psychometric properties suggest that brief versions were fine in terms of several indices of reliability (mostly internal consistency) and validity (mostly concurrent validity with other measures) and are useful as a brief screener for symptoms.

10.4.3: Single or a Few Items

Occasionally, investigators add a measure of some construct that is one or a few items. This is a case where the investigator believes it would be important to measure something but wants a very brief assessment. In the usual case, the items that are used are based on face validity only—an unacceptable indefensible criterion from the standpoint of methodology. The reason is that one does not know what any measure really assesses (with rare exception such as "death") without the proper validity data. So we may call the item, a 7-point scale to measure "empathy," for example, but without considerable other information, we have no idea what is measured. If one is tempted to use a few items that are home-made, the work behind the measure is pivotal.

Consider an example of a well-designed study focused on the antecedents of youth (ages 9–14) drinking alcohol (Fisher, Miles, Austin, Camargo Jr, & Colditz, 2007). Youth (>5,500 from different locales in the United States) were assessed and then followed for 2 years to see who would take up drinking. Among the key hypotheses was that family who ate meals together would have children who were less likely to take up alcohol use 2 years later. This was a prospective longitudinal study, so the time line (family eating meals together at time one) could be examined in relation to the outcome of interest (drinking 2 years later). Youth who ate meals with their families were much less likely to take up alcohol consumption. Yet, one always looks at the measures. How was the key construct measured? At time one, one item was used to determine if and the extent to which youth ate meals with their families. Specifically, youth were asked to answer the question: "How often do you sit down with other members of your family to eat dinner or supper?" And could answer: never, sometimes, most days, everyday. Those who answered never or sometimes were compared with those who answered most days and everyday.

The issue: A single item was used to define the construct. Although it is so tempting and appealing to be completely satisfied with face validity, that is not enough. Strictly, we have no evidence that the item measures family meal time (e.g., as opposed to social desirability as one parsimonious possibility). Another item to assess parent alcohol use was also included and that too might be explained by another construct that how frequently parents really used alcohol.

I hasten to add the use of one-item per se is not the critical point. As often as not studies use two to four items in the same vein, i.e., to mention a construct of special interest, to invent items to do that, but to assume validity and reliability without any evidence at all. Yet, assessment validation is not a luxury—we cannot rely on face validity about what an item or couple of items "really" measure. Perhaps one would argue the other way—wait, what evidence is there that the one or a few items do not measure what we say. What evidence could one use to refute in the above example that the one item measured "family eating together." Yet, this is not how science works. The onus is on us as investigators to show what a measure does assess. One cannot just say we are measuring a construct, invent a few reasonable sounding items, and ask the rest of the world to prove our measure is not valid.

The lesson from this: Use measures that go beyond face validity. One- or a few-item measures often have only that in their favor—not always. In any study, we convey to the reader (but even before that, to ourselves) that the measure we are using is a reasonable reflection of the construct we are studying. "Reasonable reflection" is a loose way of saying—is reliable and valid. A single-item or a few items might be reliable and valid, but face validity is not suitable support for either of those.

10.4.4: Considerations and Cautions

The main cautions apply primarily to shortened forms and single-or a few-item measures rather than brief forms. Again the distinction is that brief forms have as their requirements all of the usual in relation to reliability and validity and their selection ought to be based on that evidence. As a starting point, perhaps the examples (e.g., GAD 7, SCL-90 shortened forms) raise the broader issue of why do we not use brief or shortened forms for all of the lengthy measures to which we subject participants? There are methodological answers:

1. The purpose of the use of a given measure may make short forms especially helpful.

   If one wants a measure for a quick screening to identify who will be included or who will be subjected to a more intense assessment battery, short forms are particularly helpful.

   Perhaps the study only wants to include people who are experiencing mental health problems (or who are not). Use of something like an abbreviated SCL scale might have a cutoff score for the initial screening. The brief scale could be administered to many people with little inconvenience for this purpose. So this first answer pertains to the design of the study (prescreening included) and purposes of assessment.

2. There are cautions and reasons not to use short forms or shortened scales. A main reason is that the range of scores on a short version is restricted, obviously, by definition. So if one can go from 1 to 90 on a scale and 1 to 25 on a short form, that range could have implications for the study. In fact, people do not usually use the full range of a scale, so 1 to 90 probably is not accurate nor is 1 to 25 as the real range. A smaller range is likely for both long and brief forms than the numbers suggest. That means a short form is actually shorter (in range) than one might believe.

As a general rule, when we are trying to distinguish groups (e.g., group differences after an experimental manipulation or group differences after two or more groups are selected), we want the larger range of scores to help separate the groups. A restricted range makes group separation more difficult and the longer measure usually is preferred. Also, if we wish to correlate the scores of various measures, the longer scale is more likely to be beneficial because of the finer gradations (larger range) of scores that are possible. Similarly, if we want to show change (from one occasion to the next), the larger (longer) version is more likely to be able to do that because a greater range of scores means people can move on the scale in more nuanced way.

3. Related to the restricted range issue is that measures often are used to group or classify individuals. One wants high, medium, and low (e.g., sensation seeking) on some characteristic, and a measure will be used to delineate groups. Here too we want the longer form as a general rule to allow the full spread of scores and to lessen the likelihood that individuals with a score in the middle (on a shorter version) might well have been in the low or high range if more items and more nuanced scores (longer version) were available.

   Finally, shortened forms may not be appropriate if there are multiple subscales or characteristics in a measure.

   I used the SCL-90 example because research almost always uses the measure to provide an overall total symptom scores. Yet, other measures of constructs (e.g., personality, intelligence) often include subscales and separate characteristics or abilities. Here abbreviated versions may become less useful as options because items for the subscales are already relatively brief. Rather than abbreviating the measure across all items, perhaps some of the subscales could be dropped from the study because they are not relevant. That of course would abbreviate the assessment battery but retain the scales of interest.

4. A problem with short forms can be that for a given measure there may be many such forms and hence very little reliability and validity data are available for any particular form. We want to use versions of a scale that have been applied extensively and with scores showing reliabilities and validities across samples. Short forms are often evaluated but often not.

10.4.5: More Information Regarding Considerations and Cautions

Many of the above comments about restricted range can apply to brief forms, shortened forms, and single (or a few) item scales. Yet, the single- or few-item scales warrant special comment. The range on such measures is restricted so that concerns related to that are heightened. Yet, the most critical feature is the usual absence of validity about what the items measure. Outside of psychological assessment, one or two items are commonly used (e.g., answer a survey of whether you are going to purchase a new smartphone in the next 6 months, or whether you liked the experience in shopping at a Web site). Yes, these single-items have their problems, but we are talking about other issues. In science (psychological science), we are interested in constructs and their operational definitions (measures) and the extent to which those definitions reflect the constructs. We are describing phenomena, building theory, testing predictions, and so on. This is a different agenda from surveys and marketing. In conducting or reading about research psychometric properties of the scale and sensitivity reflecting change are not methodological niceties—they are essential and proper testing and support for our predictions depend on the adequacy of the measures.

What is the final word on using brief and shortened measures? There is none but critical factors to keep in mind as we make decisions in any research project or evaluating the report of someone else's research.

> Methodology (like life) often is a matter of trade-offs. The gain in brevity of a measure is being able to add it to an assessment battery when a longer version is not feasible.

Thus one has a measure of the construct and can speak to the literature that utilizes that construct and measure. The cost could be in sensitivity of the measure to show differences and change and to attenuate various statistics (e.g., correlation, effect size) because of the restricted range. In any given instance, the considerations need to be weighed (see Smith, Combs, & Pearson, 2012).

If one has to prioritize, perhaps identify the priorities of the constructs that are being evaluated in the study.

What are the main constructs to evaluate the hypotheses?

These constructs perhaps ought to be represented with more than one measure each and with as well-validated measures as available. Secondary or ancillary constructs might be better explored with brief measures. This is not a carte blanch for selecting measures with no validity data at all. Perhaps the study could provide such data. However, if the results do not "come" out with unvalidated shorted measures or the few items that are homemade, the first rival interpretation is that the measures were limited. Exert the most extreme caution in inventing a couple of items to represent a construct. That can be the worst of all worlds (no validity, restricted range, and no defensible statement to make about what was measured). When you read an article that notes something like "we used three items to measure (insert construct here)," you are entitled to roll your eyes (or someone else's eyes) if that sentence is not followed by some statement providing evidence beyond face validity.

10.5: Interrelations of Different Measures

10.5 Identify three explanations as to why the results obtained through multiple measures may vary

Although the use of multiple measures is advocated as a general strategy for evaluating a construct, this strategy has a price. The main issue stemming from use of multiple measures is that the results may be inconsistent across measures (see Meyer et al., 2001). Some dependent measures may reflect changes or differences in the predicted direction, and others may not. When results vary across measures designed to assess the same construct, this makes interpretation difficult or at least more complex.

As an example, the longitudinal study of physical abuse, sexual abuse, and neglect has shown that youths who experience abuse in childhood (<11 years old), compared with nonabused children, show greater violence and criminal behavior 20 years later (Widom & Shepard, 1996). Interestingly, the results varied by how the key predictors and outcomes were assessed. Early physical abuse in children was identified in two ways (official records, self-report in adulthood), and violence was also assessed in two ways (records of arrest and reports of violent activity). Early physical abuse, as measured by official records, predicted later arrest for violence in young adulthood. However, self-reported physical abuse did not predict later arrests. Interestingly, self-report measures of early abuse predicted self-report of violent activity. The results suggest that common method components (i.e., predictors and outcome measures that share the same methods) are related. Measures predicted less well or not at all when methods varied. The strength of the study is assessing predictors and outcomes using multiple measures and methods.

Consider another example, where newly wedded couples (N = 135) were asked to report on their partner and satisfaction with their marriage (McNulty, Olson, Meltzer, & Shaffer, 2013). There was a self-report measure that assessed their view. There was also an implicit attitude measure in which individuals responded to positive and negative words quickly presented in relation to photos of their partner or other people. The reaction time of associating positive or negative words with the photos was used to measure how positive the individual felt about the relationship or partner. Implicit measures require quick reactions and get at views people may have that are out of consciousness. In this study, there was a measure of marital satisfaction that was conscious and one that was implicit. Noteworthy is that the correlation of the two measures was 0 or written with more drama r = 0.00. As interesting, over the next 4 years, marital satisfaction was measured every 6 months. The results showed that the implicit measure predicted marital satisfaction over time whereas the self-report measure did not. One should not go too far with the conclusion. Only one self-report measure of marital satisfaction was included (and not a standardized measure of that construct) and that is a weakness before generalizing to all measures.

Even so, the lesson for this discussion is unchanged, namely, it is wise to use multiple measures to better assess a construct and to be sure or to reveal whether a given finding is not restricted to one measure.

10.5.1: Three Reasons for Lack of Correspondence among Measures

The failure of multiple measures to agree in a study is a "problem" only because of some traditional assumptions about the nature of personality and human behavior and the manner in which independent variables operate. Actually, there are many reasons to expect multiple measures not to agree. Three explanations of the lack of correspondence among measures pertain to the contribution of method variance in assessing behavior, the multifaceted nature of behavior, and the magnitude of a client's standing on the characteristic. Consider these briefly because they can help guide interpretation of a study when measures do not reflect the same pattern or results:

1. The lack of correspondence among measures of the same construct has emerged as a special topic in the context of using multiple raters or informants. For example, a couple may rate many facets of their marriage; children, parents, and teachers may rate the children's behaviors; and a patient and clinician may rate the quality of the therapy experience or therapeutic change. When different raters evaluate the "same" domain, correspondence of the separate views is often poor. For example, when multiple informants are used (e.g., children, parents, and teachers) to evaluate child behavior, the correlations of the different informants all rating the same target (child behavior) correspond between .2 and .4 (e.g., Achenbach, Krukowski, Dumenci, & Ivanova, 2005; De Los Reyes, Thomas, Goodman, & Kundey, 2013). Again it is important to be aware of the sometimes limited correspondence among informants. This is not necessarily a problem. Measures from both or all informants can be valid. Different informants sample different behaviors, experience those behaviors differently, and see individuals in different contexts (e.g., school, home, and work). As a more general issue, measures of the same construct may not invariably go together. This can occur when raters are not the "method" but different methods are used. In many ways, the absence of high agreement among measures, when it does occur, should not be surprising. Different methods of assessment present different conditions to the subjects, and these yield, generate, and promote genuinely different responses.

2. Many constructs we measure have several facets (are multidimensional) and not all of the facets may be present or go together. For example, several measures are available to assess depression. However, depression is not a unidimensional characteristic or simply sadness. Many characteristics can be identified involving affect (e.g., sadness), cognition (e.g., beliefs that things are hopeless), behavior (e.g., diminished activity), and biological symptoms (e.g., changes in eating and sleep patterns). Given the multidimensional nature of personality, behavior, and clinical dysfunction, as illustrated by depression, the lack of correspondence between measures is to be expected. Different measures of the same construct might simply emphasize or give different weight to different components, and little correspondence is quite understandable.
3. The lack of correspondence of different measures of the same construct may relate to the magnitude or strength of the characteristic, trait, disposition, or clinical problem. It may well be that different measures designed to assess the same characteristic of personality or behavior will co-vary as a function of the client's standing on the characteristic. For example, in the case of anxiety, clients may be measured on self-report, overt behavior, and psychobiological measures. The different measures may or may not correspond, depending on the magnitude of the client's anxiety. Perhaps clients who are overwhelmed by anxiety would score at a very high level on each of the measures. At the other extreme might be individuals who show absolutely no anxiety and score the equivalent of zero or "none" on each measure. These extreme groups (very high vs. no anxiety) may show a consistent or a more consistent pattern across measures.

   More generally, be alert to the prospect that multiple measures of a construct may not go together. There are ways to address this matter including combining measures when possible if they relate moderately to each other.

Also, using special statistical analyses (e.g., latent variable analysis, factor analysis) can address multiple measures of the same construct and clarify their relation to each other and to other predictors. These are beyond the present scope of the chapter but there are readily available resources (e.g., Bartholomew, Knott, & Moustaki, 2011; Duncan, Duncan, & Strycker, 2006).

10.6: Construct and Method Variance

10.6 Examine convergent and discriminant validity

It is useful to look a bit more analytically at scores on a measure. A participant's score is determined in part by standing on the construct (e.g., how depressed the person may be) and the measure that is used to assess that. The contribution of the method of assessment to the score for a given dependent measure can be seen by looking at some of the characteristics of measurement devices in general.

When a new measure is developed, it is important to establish that the measure correlates with other measures of the same construct (assuming some other measures and criteria are available) and that the measure does not correlate with measures of seemingly unrelated constructs (Campbell & Fiske, 1959).

Convergent validity, mentioned earlier (Table 10.1), was used to denote that independent methods of assessing a given construct agree with each other, i.e., correlate positively and perhaps in the moderate to high range. For example, if two measures that supposedly assess empathy correlate highly, this would be evidence of convergent validity. The fact that the measures converge suggests that they assess the same construct.

Discriminant validity was introduced to denote that a newly proposed measure should be distinguished from measures of other constructs; that is, show little or no correlation. We ought to expect a measure not to be related to other measures of entirely different constructs. Evidence that is consistent with this expectation supports discriminant validity. For example, a newly proposed measure of empathy probably would not be expected to be highly correlated with measures of other constructs, such as intelligence, social desirability, or anxiety. Evidence that all of these correlations were not very highly related to empathy would support discriminant validity of our measure. Of course, if high correlations were found between the newly proposed measure and a more established measure of another construct, this would suggest that the measures really were assessing similar characteristics, no matter what the two assessment devices were called. The new measure would be suspect to the extent that the results can be explained by a better validated measure whose construct validity (for some other construct) has been established.

The way in which convergent and discriminant validity can be examined is to conduct a study designed to evaluate the interrelations among various measures. Whether one measure converges with other measures of the same construct, of course, can be assessed by administering two or more measures designed to assess the same construct and seeing whether they correlate highly. The way to see whether a measure diverges from others with which it should not correlate is achieved by including measures of different or unrelated constructs and seeing whether they do not correlate or correlate only to a very small degree.

To obtain this set of correlations requires administering a number of measures to the same individuals. Some of the measures would be designed to assess the same construct or personality dimensions; some would assess different constructs or dimensions.

The correlations obtained to determine convergent and discriminant validity cannot be viewed uncritically. There is a

nuance here that is important to know. It is quite possible that correlations between two measures will be influenced not only by the construct that is being assessed but also by the method of assessment. For example, if two paper-and-pencil measures of empathy were administered, they might correlate rather highly. Is this evidence of convergent validity? Actually, it may be that the high correlation is due to the similarity in the method of assessment (two self-report scales) rather than or in addition to the construct being assessed. In addition, it is possible that a high correlation will be obtained between measures because of a common source of bias or artifact. For example, participants may respond in a socially desirable fashion across both measures, and this will be misinterpreted as convergent validity for the construct that the investigator originally had in mind. For the moment, the reason for the correlation between measures that use the same method is not as important as realizing that such a correlation is likely. And one criticism of a given study (that you design or read) is that all the measures were self-report. If the study is about correlations, a limit is that some of the correlations that investigator believes are very interesting might be due in part to the fact that the same assessment method was used.

10.6.1: Using a Correlation Matrix

To evaluate the contribution to the assessment method, it is important to include with the assessment of multiple constructs or dimensions some measures that rely upon different modalities or methods.

A **multitrait-multimethod matrix** refers to the set of correlations obtained from administering several measures to the same participants when these measures include more than one trait (construct) and method (Campbell & Fiske, 1959; Hox & Balluerka, 2009). The purpose of the matrix (a set of several correlations) is to evaluate convergent and discriminant validity and to examine the extent to which the correlations between measures are due to the similarities in the way the responses are assessed (method variance) rather than in what constructs supposedly are measured (trait variance).[3]

Consider in a bit more detail a real example from my own work that focused on the assessment of children hospitalized on a psychiatric inpatient unit. Two main constructs were assessed (depression and aggression). Children and mothers completed a questionnaire and an interview designed to assess depression and then a parallel questionnaire and interview designed to assess aggression. Consider this a study with two constructs (depression, aggression) and two methods of assessment (raters). All measures focused on the depression and aggression of the child.

Figure 10.1 presents the correlation matrix in the form that permits examination of convergent and discriminant validity and the role of the rater (method variance) in the

Figure 10.1: Correlations of Children and their Mothers for Measures of Depression

Correlations of Children and Their Mothers for Measures of Depression (Children's Depression Inventory [CDI] and Bellevue Index of Depression [BID]) and Aggression (Hostility Guilt Inventory [HGI] and Interview for Aggression [IA])

		Children (N = 120)				Mothers (N = 120)			
		Depression		Aggression		Depression		Aggression	
		CDI	BID	HGI	IA	CDI	BID	HGI	IA
Children[a]									
Depression	CDI								
	BID	0.62[c]							
Aggression	HGI	0.38[c]	0.42[c]						
	IA	0.41[c]	0.49[c]	0.26[b]					
Mothers									
Depression	CDI	0.10	0.24[b]	0.14	0.29[b]				
	BID	0.16[a]	0.27[c]	0.10	0.20[a]	0.71[c]			
Aggression	HGI	0.13	0.17	0.24[a]	0.24[a]	0.46[c]	0.56[c]		
	IA	0.03	−0.04	0.10	0.23[a]	0.26[b]	0.54[c]	0.60[c]	

correlations. First, several numbers are in diagonals (and not enclosed within a triangle). These are correlations when the rater is different but the construct and measure are the same. These are used to support convergent validity. (Because these measures were administered at a given point in time, these also happen to be measures of concurrent validity.) Look at the circled numbers in the first column (vertical) in the table. For child and mother scores for the Children's Depression Inventory the correlation (r) is .10. This of course means that the children and mothers do not agree very much at all on the level of the children's depression. Disappointing but not odd; we have known for a while that parent–child agreement on child symptoms is very low.

The solid-line triangles include correlations of measures completed by the same rater (children or mothers). Within the triangles, it is useful to look at the correlation is with the same rater but different constructs. The dashed-line triangles include correlations of measures completed by different raters. Together, the different triangles show there is a rater effect, i.e., correlations tend to be higher when the rater is the same no matter what construct was measured. The diagonal rows of numbers between the dashed line triangles are validity correlations (completion of the same measures by different raters) ap< .05, bp< .01, cp< 001.

Of greater interest are the correlations enclosed in the triangles. The solid triangles include measures completed by the same rater (child or mother). In general, the correlations within a given solid triangle correlate relatively highly. This means that measures completed by the same rater or informant share an important source of variance even if the constructs they rate are quite different. Indeed, correlations by the same rater who rates different constructs (depression, aggression) tend to be higher than correlations of the same construct (depression) by different raters. For example, child ratings of depression and aggression on two interviews (BID, IA in the Table) were correlated at r = .49 (same rater and measures but different constructs). This is higher than ratings of depression between mother and child which were correlated at rs = .10 and .27 (same construct different raters).

The example permits separation of construct (trait variance) from rater (method variance). The study and many others like it show the strong contribution of rater variance. That is, measures of different traits within raters often are more likely to correspond than measures of the same traits between different raters. This finding has been consistent across different informants, including children, parents, teachers, peers, and hospital staff as sources of information (e.g., De Los Reyes et al., 2013). Stated more succinctly, there is a strong method (rater) component that often pervades the ratings, even though children can be reliably distinguished in terms of their depression and aggression.

Assessment in experiments can profit from knowledge that both substantive and methodological characteristics of assessment devices contribute to a subject's score. If information is desired about a construct, it is important to use more than a single measure. Any single measure includes unique components of assessment that can be attributed to methodological factors.

> Evidence for a particular hypothesis obtained on more than one measure increases the confidence that the construct of interest has been assessed.

The confidence is bolstered further to the extent that the methods of assessing the construct differ.

10.7: General Comments

10.7 Review the need to ensure that the selected measure assesses the construct of interest

The discussion of special issues related to selecting measures can be highlighted by noting several questions that can serve as guides in designing a study or evaluating one that has been completed.

> **First**, is there evidence that the measure one has selected actually assesses the construct of interest?
>
> The measure may seem reasonable (face validity) or someone may have used it before (argument by authority). These reasons are not enough for psychological (scientific) assessment. We need more and that is what the various forms of reliability and validity address.
>
> **Second**, what are possible factors that influence people's scores on the measures, and could these affect our conclusions?
>
> Three factors that come to mind are response sets or styles (social desirability), conditions of assessment (reactivity), and unique features associated with the assessment method (e.g., self-report). These can be surmounted by using more than one measure and having those measures vary in the assessment method (self-report and overt behavior; or neuroimaging and other-report) and if possible the conditions of assessment (e.g., unobtrusive measures).
>
> **Third**, could method factors limit the conclusions of a study?
>
> Yes, if multiple constructs are assessed with the same method, any correlations between the measures could be heavily influenced by common method factors. For example, much research now is conducted via the Web (e.g., MTurk). Participants are asked to fill out a set of self-report measures of different constructs (e.g., depression, relationship satisfaction, views of this or that social

issue). And now the investigator correlates all of these. This is not a strong study in part because the correlation includes the relation of the constructs to each other but also the common method factor—all self-report. The correlations may be inflated because of the shared method factor that pervades all of the measures.

Fourth, is there a better way to address many of the issues? Yes, for a key construct in a given study (e.g., aggression, arousal), try to use two or more measures that use different methods. We want to know whenever we can whether the results of our study are restricted to a single measure (narrow stimulus sampling as a threat to external and construct validity). Measuring the construct in more than one way adds strength to a study. If the results are consistent across the measures, that very much adds weight to the conclusions. If the results are not consistent across measures, that suggests intriguing hypotheses and prompts a little theory as to what happened. That too is an important contribution.

Summary and Conclusions: Selecting Measures for Research

Selection of measures for research is based on several considerations, including construct validity, psychometric properties, and sensitivity of the measures to reflect changes or differences. Also, it is important to consider the sample for which the measure will be used and whether psychometric properties apply to the use you intend. Culture and ethnicity were discussed. As measures are used with diverse groups, it cannot be assumed that the measures have the same meaning or properties. It is critically important to consider individual measures whether they are best suited to the goals of the project you are designing. Psychometric properties from prior studies as relevant to what you will be doing (e.g., test–retest reliability of repeated testing in your study) are a guide. This stands in contrast to a frequent way in which a given measure is—it was used before by someone else. There are better reasons to use a measure or to do almost anything else in life than, "someone else did it before."

Standard or currently available measures are usually used in a given study because they have considerable evidence in their behalf and because as investigators we wish to have our findings relevant to a broader literature in which the measures are frequently used. Occasionally, investigators alter standardized measures to apply them to populations or in contexts in which the measures have not been used or intended. The measure may be used as is or modified slightly by rewording or omitting items. Investigators may develop an entirely new measure because a standard measure is not available or alteration of an existing measure would on prima facie grounds render this of limited value. If a measure is used in a novel way, altered in any way, or if a new measure is developed, it is essential to include validity data within the study or as pilot work to that study to support construct validity.

Special issues were discussed that also guide selection of measures. Awareness of being assessed was discussed and may be a common method factor across all measures within a study and influence the findings. Socially desirable responding was one influence that may be prompted by being aware that one is being assessed.

Use of multiple measures rather than a single measure was recommended because:

- Constructs of interest (e.g., clinical problems, personality, social functioning) tend to be multifaceted, and there is no single measure that can be expected to address all of the components
- Performance may vary as a function of the assessment modalities and devices used
- Individual's standing on a particular dimension or construct may be partially influenced by the method of assessment

In advocating use of multiple measures, the assessment battery can be prohibitive for the participant. Hours (days) may be required to complete the assessment battery. Brief, shortened, and single-item measures were discussed. Many cautions were presented too because the primary criteria to keep in mind in measurement selection were those outlined at the beginning of the chapter (evidence for construct validity, psychometric properties, and measurement sensitivity). Shortened measures and the single- or few-item measures often do not have the requisite data to recommend their use or to allow their interpretation.

The interrelation of multiple measures of the same construct was also discussed. Different measures of the same construct do not always go together. This has been well-studied in the context of multiple informants who evaluate the "same" target (e.g., their marriage, child,

"life"). There are reasons to expect measures will not invariably go together as they sample different facets of a construct or different perspectives.

Finally, construct and method variance were discussed. The purpose was to underscore the fact that the method contributes or can contribute to the results. Also, a common method factor (e.g., all self-report scales) can inflate relations among measures. Convergent and discriminant validity and the multitrait-multimethod matrix help to reveal the contributions of construct and method variance and to provide an example of how they can be separated. Also, the fact that method of assessment can contribute to the results argues for using multiple measures that vary in the method of assessment.

This chapter has focused on the considerations underlying measurement selection. Among the common themes was considering the use of multiple measures to operationalize a construct and to use measures of different methods when possible. The next chapter focuses on the different types of measures from which to use and special issues that each type of measure may raise.

Critical Thinking Questions

1. Think of a situation in which some characteristic is measured and the measure, situation, or circumstances are likely to be reactive, i.e., change how a person responds from what it might otherwise be. How might you measure that same characteristic that would minimize reactivity?
2. You have developed a measure of "courage," which is designed to reflect how brave a person is in difficult situations. Think about validity in the following ways. What other characteristics would courage be related to (concurrent validity, convergent validity) if a study were done looking at different personal characteristics? What other characteristics would you expect to NOT relate to courage at all or very little (discriminant validity)? The answers to both of these questions contribute to the construct validity of your measure.
3. What could be misleading by using measures that only have face validity in their behalf?

Chapter 10 Quiz: Selecting Measures for Research